
SCI-RTC

Raw-Timestream Conditioning and Temporal Response

Engineering Conformance View

Package version	v0.1
Revision	r0.2
Date	2026-08-18
Authority	Shared normative core in <code>src/common/</code>
View-specific content	Informative evidence and review guidance only
Status	Implementation-blind author draft for scientific-owner review

Claim boundary. This view does not assert scientific approval, implementation conformity, representation fidelity, validation, observational performance, or production readiness. It contains no independent normative science.

How to use this view

The shared normative core below is byte-for-byte the same set of six inputs used by the scientist-facing rationale and is imported exactly once. It is the only authority in this PDF. The material after the end-of-core marker is an informative method for recording evidence against that authority. It may not waive a requirement, select an owner decision, redefine an unavailable state, or restate an equation with different semantics.

A conformance review should create one evidence record per requirement and one falsification record per applicable prediction. Each record must bind an exact implementation and artifact identity, state what was inspected or executed, preserve unexpected error-level messages, and distinguish pass, fail, not exercised, blocked by owner decision, and unavailable evidence. Passing a fixture supports only the named requirement and claim layer.

Stop rule. If a required owner decision is open, the affected operation or claim remains unavailable. Engineering must not choose a numerical value from current behavior and then describe it as scientific authority.

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1 Shared normative authority and notation

Normative core stamp: SCI-RTC-v0.1/r0.2. The six files in `src/common/` form one authority. Text outside this shared core is explanatory or conformance guidance and cannot alter a definition, equation, assumption, requirement, prediction, unavailable state, or owner decision.

SCI-RTC begins with an admitted aligned primary detector stream and ends with an atomic, role-specific conditioning bundle. ALIGN, AST, SCI-BEAM, and SCI-CAL retain authority for the upstream meanings they produce. PTC, VAL, MAP, BEAM, NOI, MAP-003, FLT, SRC/MODE, and FRUIT retain authority for their downstream science. This contract owns the realized conditioning order and its scientific consequences; it does not infer current software behavior.

The words **shall** and **shall not** identify requirements. **Required unavailable** means that the affected product or claim shall not be reported as complete. **Optional unavailable** is a typed, scientifically inert state. Missing, disabled, automatic, rejected, invalid, failed, unavailable, unsupported, and omitted remain distinct states.

1.1 Indices, sets, and role labels

Symbol	Normative meaning
o	Immutable observation identity.
$d, q \in \mathcal{D}_o$	Exact admitted detector occurrences; d is a target and q may be a donor. A detector index or row number alone is not this identity.
r	Native acquisition row; it is not an aligned or conditioned slot.
$j \in \{0, \dots, N-1\}$	Ordered aligned input slot on the admitted ALIGN grid.
$n \in \{0, \dots, N_y-1\}$	Ordered phase-zero RTC output slot.
$\mathcal{R}$	Product role, including Pointing, OOF, Science, Beammap, diagnostic-only, or another separately approved role.
$\delta_{\mathcal{R}}$	Role-domain selector: 0 for an admitted raw role and 1 for a separately CAL-authorized calibrated role.
x_{dj}^A	Admitted aligned primary xs value with declared physical observable, sign, baseline/reference, unit, support, and validity.
z_{dj}	Calibrated value when the role admits an imported CAL operator.
u_{dj}	Role-domain value immediately before the ordered RTC filter chain.
y_{dn}	Conditioned role-domain output at the selected phase-zero slot.
ϕ_d	flxscale for one exact detector occurrence under the declared multiplicative convention and calibration domain.
A_{α}	Imported, realized ALIGN mapping with state α ; its scientific meaning is not owned by RTC.
C_{κ}	Imported CAL operator with state κ ; it is the identity for a raw role.
B_{ω}	Realized despike/replacement operator conditioned on selection state ω .
F_{k, θ_k}	Realized ordered FIR, IIR, notch, or admitted mask-controlled stage k .
$D_{M,0}$	Integer phase-zero point selector of factor $M \geq 1$.
Ω	Complete realized conditioning state, including all upstream bindings and every selected stage, exception, state, edge rule, and output interval.
$\mathcal{L}$	Declared learning population with exact identity, selection, support, masks, exclusions, and compatibility domain.
Λ	Learned evidence and uncertainty produced by declared diagnostics; it is not authorization to apply.
Π	One immutable resolved RTC plan admitted from learned evidence and fixed scientific policy.
$\mathcal{A}_{\Pi}$	Apply operator that consumes plan Π without changing it.
$K_{dn,qj}$	Local detector-time response of one output cell to one acquired input cell.
$\mathcal{S}_{dn}$	Exact transitive input support of output (d, n) , including alignment, donors, filters, state, and selection.
$\mathcal{I}_{dn}$	Cause-preserving transitive influence set for output (d, n) .

1.2 Identity and interval conventions

Observation, Tune, array, network/interface, detector occurrence, selected APT row/binding, native row, aligned slot, scan, science interval, context interval, conditioned output slot, stage, and product identities are non-interchangeable. Array identity is one of **a1100**, **a1400**, and **a2000**; array ID, array index, network ID, detector index, and container position do not substitute for one another. Cross-artifact identity requires an explicit occurrence- and product-scoped binding. Equal row numbers do not establish identity.

Primary detector timestreams are samples on rows and detectors on columns unless a product explicitly declares another ordered shape. Every interval is half-open. Science windows, physical scans, processing chunks, filter context, output selections, and learned-plan scan sets retain separate identities. Native, aligned, and output indices never silently renumber one another.

Time is seconds on the admitted ALIGN-assigned grid. The approximately 8-ms rate relationships and the 0.5 $\times$, 2 $\times$, and 4 $\times$ family are conditional software-grid facts, not proof of a detector integration event, absolute epoch, or astrometric correction. Coordinate-dependent Pointing, OOF, and Beammap controls use a declared AltAz tangent plane; Science controls use a declared equatorial J2000 TAN relation. RTC performs no implicit frame conversion.

1.3 Logical and availability notation

$\mathbb{I}[\cdot]$ denotes an indicator, $\mathbb{E}$ expectation, Cov covariance, and $\text{supp}(\cdot)$ operator support. The logical symbols $\wedge$, $\vee$, and $\neg$ have their usual meanings. A quantity decorated available is present with its required identity, domain, and validity; a quantity decorated unavailable is explicitly unavailable. Neither zero nor NaN is an availability encoding.

End of notation; shared definitions follow without a change of authority.

2 Normative definitions

SCI-RTC-DEF-001 – Admitted aligned stream.

An immutable observation-scoped primary **xs** array whose physical observable, sign, zero/reference, baseline, unit, ordered detector binding, assigned sample grid, cadence, phase, support, origin/synthesis state, validity, scan partition, and mapping lineage are supplied by the responsible upstream authority. Admission does not upgrade unavailable physical event timing.

SCI-RTC-DEF-002 – Product-role signal domain.

The tuple $(\mathcal{R}, Q_{\mathcal{R}}, U_{\mathcal{R}}, s_{\mathcal{R}}, b_{\mathcal{R}}, V_{\mathcal{R}})$ naming role, physical quantity, unit, sign, reference/baseline, and valid domain. Frozen Beammap uses raw $\Delta f/f$. A separately CAL-authorized role may use calibrated mJy/beam. The domains never substitute silently.

SCI-RTC-DEF-003 – Exact detector occurrence.

An observation-local layered binding containing observation, Tune, network/interface, network-local tone or admitted row slot, selected APT identity/digest, and row association. A transient column number is only a locator.

SCI-RTC-DEF-004 – Imported CAL operator.

An immutable SCI-CAL-selected transformation carrying its detector binding, source and target atmosphere lineage, target unit, factor convention, validity, uncertainty, availability, and provenance. RTC may apply it in an authorized role and owns that order; RTC does not derive, repair, or promote the factor.

SCI-RTC-DEF-005 – Compatible flxscale pair.

Two finite factors ϕ_q and ϕ_d that are valid for the exact donor and target detector occurrences under the same multiplicative calibration convention, quantity domain, reference plane, and applicable support, with $\phi_d \neq 0$. Compatibility is not established by matching row numbers or finite values alone.

SCI-RTC-DEF-006 – Selected conditioning policy.

A versioned, admitted specification of despike detection, donor eligibility, donor scoring and ties, replacement and fallback, source protection, filter stages and order, exact coefficients and normalization, numeric precision, state/reset, boundaries/edges, non-finite handling, sampling mode, phase-zero factor, flags, influence aggregation, diagnostics, and failure behavior. It is selected authority, not an implementation default.

SCI-RTC-DEF-007 – One-way state record.

The ordered record $(\Theta^{\text{req}}, \Theta^{\text{eff}}, \Theta^{\text{obs}}, \Theta^{\text{learned}}, \Theta^{\text{resolved}}, \Theta^{\text{real}})$. Requested state preserves the accepted request; effective state is the admitted typed plan; observation-resolved state binds exact observation facts; learned state records evidence when learning occurs; resolved state is the immutable admitted plan for every fixed or learned operation; and realized state records what ran. A later state never rewrites an earlier state.

SCI-RTC-DEF-008 – Realized conditioning operator.

The chronological composition of every admitted alignment, role-domain, replacement, mask, FIR/IIR/notch, boundary, state, non-finite, and phase-zero sampling action, including all affine state terms and exception intervals. Annotations that affect values, support, influence, or availability are part of the operator.

SCI-RTC-DEF-009 – Phase-zero sampling.

Point selection of the pre-sampling stream at input indices $j = Mn$ for one integer factor $M \geq 1$. It is not an arithmetic mean, block aggregate, or implicit anti-alias filter.

SCI-RTC-DEF-010 – Complete realized response.

An exact local or factorized representation that includes every enabled response-changing stage, detector mixing,

mask, state, edge rule, exception, and sampling phase on its declared support. If such a representation is not truthfully available, response state is unavailable; a partial kernel is not complete response.

SCI-RTC-DEF-011 – Direct source cell.

An aligned cell that enters an RTC value before transitive propagation. It may be original, ALIGN-synthesized, or RTC-replaced. Numerical participation does not confer independent scientific exposure.

SCI-RTC-DEF-012 – Transitive influence.

The cause-preserving closure from every original, synthesized, replaced, masked, edge, state, and non-finite source cell through all nonzero realized alignment, donor, filter, state, and sampling dependencies to an output. Influence is broader than the selected point and may cross detector columns.

SCI-RTC-DEF-013 – Scientifically ineligible RTC output.

An output having any transitive influence from an ALIGN-synthesized or RTC-replaced cell. It remains ineligible even when finite and even when the influenced cell is not the phase-zero selected center. Causes and numerical influence remain reportable.

SCI-RTC-DEF-014 – Coordinate-dependent control.

A source mask or response operation that consumes an AST coordinate identity, frame, topology, detector binding, validity, uncertainty where required, and support. Invalid or unavailable coordinates are not outside-source values.

SCI-RTC-DEF-015 – Fixed sampling mode.

A request whose admitted factor, realizable prefilter, rate, phase-zero rule, and downstream compatibility are resolved without learning them from the observation.

SCI-RTC-DEF-016 – Learned sampling mode.

An optional request that evaluates an admitted finite candidate set and resolves one immutable observation-wide cadence/filter plan before apply. Its admission uses the smallest admitted beam, maximum valid in-scan speed, analytical beam-times-realized-filter response, exact phase-zero alias accounting, sampling sufficiency, filter realizability, and downstream compatibility. Noise-aware, per-array, per-scan, heterogeneous-cadence, and continuous adaptation are outside v0.1.

SCI-RTC-DEF-017 – Conditional statistical covariance.

The covariance obtained from an admitted input covariance through a fixed realized operator while holding selection and nuisance state fixed. It is not total uncertainty and does not imply a diagonal, white, stationary, or independent noise model.

SCI-RTC-DEF-018 – Atomic RTC bundle.

One role-specific completion unit containing conditioned TOD; ordered detector/sample/scan/support and parent identity; complete response or typed unavailability; cause-preserving flags and influence; separated validity and eligibility inputs; conditional uncertainty/covariance status; one-way state and provenance; required diagnostics; and completion/failure state. TOD alone is not this bundle.

SCI-RTC-DEF-019 – Scientifically named diagnostic.

A quantity with an explicit estimator, unit, support, selection/mask, validity, uncertainty/availability, and policy role. It is either inert, advisory, or an input to an explicitly selected policy; those roles are not interchangeable.

SCI-RTC-DEF-020 – Claim layer.

One of algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, or production readiness. Evidence at one layer does not establish another.

SCI-RTC-DEF-021 – Declared learning population.

The exact training interval, designated same-observation interval, scan collection, observation aggregate, or other named population used to infer an RTC quantity, including identity, support, selection, masks, exclusions, signal role, compatibility domain, and completeness. A population is not defined by whatever samples happen to be available during execution.

SCI-RTC-DEF-022 – Learned evidence.

The output of named diagnostics applied to a declared learning population, including candidate parameters, estimator, unit and convention, support, quality, uncertainty, validity, and failure state. Learned evidence proposes or constrains a plan; it does not authorize apply.

SCI-RTC-DEF-023 – Resolved RTC plan.

One immutable observation- and role-bound specification admitted only after all required scientific predicates pass. It contains operation sequence, family, exact coefficients or identity, rates and conventions, order/taps, bands and tolerances, notch state, applicability, masks, donor policy, edges, guards, resets, decimation and timing, expected response/support, uncertainty, validity, and downstream compatibility.

SCI-RTC-DEF-024 – Apply state.

Execution of exactly one resolved plan on compatible admitted input. Apply records actual intervals, exceptions, resets, outputs, and completion but does not relearn, retune, change policy, or rewrite the resolved plan.

SCI-RTC-DEF-025 – Online-adaptive estimator.

A separate time-varying, data-dependent estimator whose coefficients or other operator parameters evolve during apply under a declared update rule. Its learning support, stability, response, uncertainty, guard, reset, and restart semantics are not those of learned-then-fixed filtering. It is unavailable in v0.1 absent successor authority.

SCI-RTC-DEF-026 – Scientific filter design specification.

The physical purpose, source/contaminant model, admitted scan/beam/source domain, pass/transition/stop or notch constraints, phase and registration, alias bound, support/guard, numeric precision, realizability, validation, and failure criteria that justify a filter family and its numerical parameters. An operation label or coefficient array alone is not this specification.

3 Normative equations and identities

The equations in this section are the sole displayed mathematical authority of both views. They describe an implementation-independent, state-conditioned operator. They neither select numerical policy nor assert implementation conformity.

3.1 Role domain and donor transfer

For an exact detector occurrence i under the admitted multiplicative raw-to-calibrated convention,

$$z_i = \phi_i x_i, \quad \phi_i = \text{flxscale}_i. \quad (\text{SCI-RTC-EQ-001})$$

Equation (SCI-RTC-EQ-001) does not assign a role to legacy **responsivity**. When q is a raw donor and d is a raw target, the unique multiplicative transfer that represents donor-calibrated amplitude in the target raw convention is

$$x_{d \leftarrow q}^R = a_{d \leftarrow q} x_q, \quad a_{d \leftarrow q} = \frac{\phi_q}{\phi_d}, \quad \phi_d x_{d \leftarrow q}^R = \phi_q x_q. \quad (\text{SCI-RTC-EQ-002})$$

This equation is defined only for a compatible **flxscale** pair as defined above. Otherwise this transfer is unavailable. An additive calibration convention, a different unit/reference domain, or an alternative donor authority requires its own complete affine equivalence and is not silently folded into equation (SCI-RTC-EQ-002).

Let the role-domain operator be

$$C_{\mathcal{R}} = \begin{cases} I, & \delta_{\mathcal{R}} = 0 \quad (\text{admitted raw role}), \\ C_{\kappa}, & \delta_{\mathcal{R}} = 1 \quad (\text{CAL-authorized role}). \end{cases} \quad (\text{SCI-RTC-EQ-003})$$

For a CAL-authorized role, calibration precedes cross-detector replacement. For a raw role, $C_{\mathcal{R}} = I$ and an admitted raw donor transfer may use equation (SCI-RTC-EQ-002). Reversing calibration and replacement is equivalent only if, for the complete admitted affine operators on the exact support,

$$B_{\omega}^{(z)} C_{\kappa} = C_{\kappa} B_{\omega}^{(x)}, \quad \mathbf{b}_{\omega}^{(z)} = C_{\kappa} \mathbf{b}_{\omega}^{(x)} + \mathbf{c}_{\kappa} - B_{\omega}^{(z)} \mathbf{c}_{\kappa}, \quad (\text{SCI-RTC-EQ-004})$$

with unit, factor direction, additive offset, response, uncertainty, identity, validity, and domain equivalence all proved. Equality of numerical values in one fixture is insufficient.

3.2 Realized conditioning operator

Conditioned on complete realized state Ω , the ordered role-specific operator is

$$\mathbf{y}_{\mathcal{R}} = D_{M,0} F_{K,\theta_K} \cdots F_{1,\theta_1} B_{\omega,\mathcal{R}} [C_{\mathcal{R}} (A_{\alpha} \mathbf{x}^{\text{acq}} + \mathbf{a}_{\alpha}) + \mathbf{c}_{\mathcal{R}}] + \mathbf{b}_{\Omega}. \quad (\text{SCI-RTC-EQ-005})$$

Chronological order is read from right to left. The affine terms include only declared calibration, replacement, filter-state, boundary, or other authorized contributions. Masks, coefficients, finite handling, context, exceptions, and state are components of Ω , not annotations outside the operator. Conditional on them,

$$\mathbf{y}_{\mathcal{R}} = L_{\Omega} \mathbf{x}^{\text{acq}} + \mathbf{b}'_{\Omega}, \quad L_{\Omega} = D_{M,0} F_K \cdots F_1 B_{\omega,\mathcal{R}} C_{\mathcal{R}} A_{\alpha}. \quad (\text{SCI-RTC-EQ-006})$$

This is a conditional affine identity, not an unconditional linear, stationary, unbiased, or Gaussian claim.

A realized replacement at target cell (d, j) may be written

$$u_{dj}^R = (1 - p_{dj})u_{dj} + p_{dj} \left[\sum_{(q, \ell) \in \mathcal{Q}_{dj}} w_{dj, q\ell} a_{d \leftarrow q, j\ell} u_{q\ell} + \beta_{dj} \right], \quad \sum_{q, \ell} w_{dj, q\ell} = 1 \text{ when normalization is claimed.} \quad (\text{SCI-RTC-EQ-007})$$

The scale a is domain-qualified: equation (SCI-RTC-EQ-002) is the authorized raw `flxscale` route, while a calibrated-domain transfer or any separately authorized route carries its own identity and validity.

For a time-invariant FIR stage,

$$v_{d,j} = \sum_{\ell \in \mathcal{L}_k} h_{d,\ell}^{(k)} u_{d,j-\ell}, \quad H_d^{(k)}(e^{i\omega}) = \sum_{\ell \in \mathcal{L}_k} h_{d,\ell}^{(k)} e^{-i\omega\ell}, \quad H_d^{(k)}(1) = \sum_{\ell} h_{d,\ell}^{(k)}. \quad (\text{SCI-RTC-EQ-008})$$

For a causal IIR or notch stage, a state-complete form is

$$\mathbf{s}_{j+1} = A_k \mathbf{s}_j + B_k u_j, \quad v_j = C_k \mathbf{s}_j + D_k u_j, \quad \mathbf{s}_{j_0} = \mathbf{s}_{0,k}. \quad (\text{SCI-RTC-EQ-009})$$

An exactly equivalent recurrence is admissible only with its coefficient convention, precision, state, and boundary identity. A causal-stability claim requires every realized pole strictly inside the unit circle.

One possible selected mask operator is normalized masked FIR convolution,

$$v_j = \frac{\sum_{\ell} h_{\ell} m_{j-\ell} u_{j-\ell}}{\sum_{\ell} h_{\ell} m_{j-\ell}}, \quad \left| \sum_{\ell} h_{\ell} m_{j-\ell} \right| > \epsilon_m, \quad (\text{SCI-RTC-EQ-010})$$

but equation (SCI-RTC-EQ-010) is not a selected default. Segmenting, zero filling, restoring a mask after filtering, holding state, and renormalizing are distinct operators.

3.3 Phase-zero sampling and response

For a coherent pre-sampling segment of length N , v0.1 selects only

$$y_{d,n} = u_{d,Mn}, \quad 0 \leq Mn < N, \quad f_{\text{out}} = \frac{f_{\text{in}}}{M}, \quad N_y = \begin{cases} 0, & N = 0, \\ 1 + \left\lfloor \frac{N-1}{M} \right\rfloor, & N > 0. \end{cases} \quad (\text{SCI-RTC-EQ-011})$$

The representative assigned-grid time is the selected point time $t_n^{\text{out}} = t_{Mn}$; the scientific support is nevertheless the complete transitive support $\mathcal{S}_{dn}$ of all preceding stages. A selected edge/output policy may reject candidate points, but shall not relabel a different phase or silently change equation (SCI-RTC-EQ-011).

For output (d, n) and acquired input (q, j) , the exact local response is

$$K_{dn,qj} = \frac{\partial y_{dn}}{\partial x_{qj}^{\text{acq}}} \quad \text{at fixed realized state } \Omega. \quad (\text{SCI-RTC-EQ-012})$$

At a selection boundary the derivative may be discontinuous or undefined; that state shall be represented rather than replaced by an LTI claim.

In the fixed, detector-separable, translation-invariant, single-rate interior case only,

$$H_d^{\text{comb}}(e^{i\omega}) = \prod_{k=1}^K H_d^{(k)}(e^{i\omega}), \quad \tau_g(\omega) = -\Delta t \frac{d}{d\omega} \text{unwrap}[\arg H_d^{\text{comb}}(e^{i\omega})]. \quad (\text{SCI-RTC-EQ-013})$$

Amplitude is $|H|$, power response is $|H|^2$, and phase is $\arg H$ under the recorded Fourier convention. Group delay is unavailable at response zeros unless a bounded convention is supplied.

For an LTI prefilter followed by phase-zero selection, the exact multirate identity is

$$Y(e^{i\Omega_o}) = \frac{1}{M} \sum_{r=0}^{M-1} H(e^{i(\Omega_o + 2\pi r)/M}) X(e^{i(\Omega_o + 2\pi r)/M}), \quad (\text{SCI-RTC-EQ-014})$$

under the declared Fourier/sign convention. Every folded image contributes; point selection itself is not anti-alias filtering.

For an FIR output, direct support starts with

$$\mathcal{S}_{dn}^{(1)} = \{Mn - \ell : \ell \in \mathcal{L}\}, \quad \mathcal{S}_{dn} = \text{closure}_{A,B,F,\text{state}}\left(\mathcal{S}_{dn}^{(1)}\right). \quad (\text{SCI-RTC-EQ-015})$$

IIR support extends to the declared state/reset or context boundary and may be mathematically infinite. Any finite guard is an approximation with a stated residual-tail criterion.

3.4 Moments, influence, and learned-plan selection

For admitted input mean $\boldsymbol{\mu}_x$ and full covariance Σ_x , conditional on fixed Ω independent of the modeled random noise,

$$\mathbb{E}[\mathbf{y} \mid \Omega] = L_\Omega \boldsymbol{\mu}_x + \mathbf{b}'_\Omega, \quad (\text{SCI-RTC-EQ-016a})$$

$$\Sigma_{y|\Omega}^{\text{stat}} = L_\Omega \Sigma_x L_\Omega^\top. \quad (\text{SCI-RTC-EQ-016b})$$

If selectors depend on the random data, unconditional covariance obeys

$$\text{Cov}(\mathbf{y}) = \mathbb{E}_\Omega[\text{Cov}(\mathbf{y} \mid \Omega)] + \text{Cov}_\Omega(\mathbb{E}[\mathbf{y} \mid \Omega]). \quad (\text{SCI-RTC-EQ-017})$$

The second term is not zero by default. With named nuisance parameters $\boldsymbol{\eta}$ and an adequate local linearization,

$$\Sigma_y^{\text{sys}} \approx J_\eta \Sigma_\eta J_\eta^\top, \quad J_\eta = \left. \frac{\partial \mathbb{E}[\mathbf{y} \mid \Omega, \boldsymbol{\eta}]}{\partial \boldsymbol{\eta}} \right|_{\boldsymbol{\eta}_0}, \quad (\text{SCI-RTC-EQ-018})$$

and a total-covariance claim requires every applicable component and cross term:

$$\Sigma_y^{\text{total}} = \Sigma_y^{\text{stat}} + \Sigma_y^{\text{sys}} + \Sigma_y^{\text{selection}} + \Sigma_y^{\text{model}} + \Sigma_y^{\text{cross}}. \quad (\text{SCI-RTC-EQ-019})$$

Unknown components are unavailable, not zero.

Define cause-preserving influence and the v0.1 ineligibility rule by

$$\mathcal{I}_{dn} = \text{closure}_{K,\text{generative links}}\{\text{typed source causes on } \mathcal{S}_{dn}\}, \quad (\text{SCI-RTC-EQ-020a})$$

$$E_{dn}^{\text{RTC}} = V_d^{\text{det}} \wedge V_{dn}^{\text{num}} \wedge V_{dn}^{\text{support}} \wedge V_{dn}^{\text{response}} \wedge \neg \mathbb{I}[\text{ALIGN_synth} \in \mathcal{I}_{dn}] \wedge \neg \mathbb{I}[\text{RTC_replace} \in \mathcal{I}_{dn}]. \quad (\text{SCI-RTC-EQ-020b})$$

Consumer-specific coordinate or policy predicates may further restrict eligibility; they may not remove either transitive-influence exclusion.

For learned mode, let $\mathcal{M}$ be the admitted finite integer candidate set and $\mathcal{F}(M)$ the conjunction of all owner-resolved astronomical transfer, exact phase-zero alias, sampling, realized-filter realizability, and downstream-compatibility constraints, evaluated with the smallest admitted beam and maximum valid in-scan speed. Then

$$\mathcal{M}_{\text{safe}} = \{M \in \mathcal{M} : \mathcal{F}(M) = \text{true}\}, \quad M_\star = \max \mathcal{M}_{\text{safe}}. \quad (\text{SCI-RTC-EQ-021})$$

Equation (SCI-RTC-EQ-021) is defined only after every numerical tolerance, support rule, fallback, and compatibility predicate named in the owner ledger is resolved. Percentile speed is diagnostic, not safety authority.

Finally, the atomic product is

$$\mathcal{O}_{\text{RTC},\mathcal{R}} = (\mathbf{y}, \mathcal{J}, \mathcal{K}, \mathcal{S}, \mathcal{I}, \mathcal{F}, \mathcal{V}, \mathcal{U}, \mathcal{P}, \mathcal{D}, \mathcal{C}), \quad (\text{SCI-RTC-EQ-022})$$

where $\mathcal{J}$ is ordered identity, $\mathcal{K}$ complete response status, $\mathcal{S}$ support, $\mathcal{I}$ influence, $\mathcal{F}$ typed flags/causes, $\mathcal{V}$ separated validity and eligibility inputs, $\mathcal{U}$ uncertainty/covariance availability, $\mathcal{P}$ one-way state/provenance, $\mathcal{D}$ diagnostics, and $\mathcal{C}$ atomic completion or required-output failure state.

3.5 Scientific design and learn–resolve–apply identities

For a locally Gaussian effective beam with angular covariance Σ and angular scan-velocity vector $\mathbf{v}$, the projected source-crossing time scale is

$$\sigma_t = (\mathbf{v}^\top \Sigma^{-1} \mathbf{v})^{-1/2}. \quad (\text{SCI-RTC-EQ-023})$$

Equation (SCI-RTC-EQ-023) is a local Gaussian diagnostic. Scientific admission shall use the exact source-times-realized-response model over the declared beam, scan, source-class, and cadence domain.

A generic realized notch may be represented, where applicable, by

$$H_{\text{notch}}(z) = g \frac{\prod_{a=1}^{N_z} (1 - z_a z^{-1})}{\prod_{b=1}^{N_p} (1 - p_b z^{-1})}, \quad |p_b| < 1 \text{ for a causal-stability claim.} \quad (\text{SCI-RTC-EQ-024})$$

This form imposes no universal biquad. Exact realized coefficients, sample rate, frequency convention, phase/direction, state, edges, and precision are part of the response.

For an admitted FIR design family H_N , a selected order shall follow a declared constrained problem of the form

$$N_\star = \min_N \left\{ N : \begin{array}{l} \sup_{f \in \mathcal{P}} |H_N(f) - H_{\text{target}}(f)| \leq \epsilon_{\mathcal{P}}, \\ \sup_{f \in \mathcal{S}} |H_N(f)| \leq \epsilon_{\mathcal{S}}, \\ \text{phase/delay, alias, edge/guard, precision, and realizability predicates pass} \end{array} \right\}. \quad (\text{SCI-RTC-EQ-025})$$

The sets, metrics, tolerances, equality conventions, and family are owner-resolved; an approximate order formula is not universal authority.

For a causal linear-phase FIR with N taps at rate f_{samp} , its nominal interior group delay is

$$\tau_g = \frac{N - 1}{2f_{\text{samp}}}. \quad (\text{SCI-RTC-EQ-026})$$

Where a temporal response has effective delay τ during a locally uniform scan, the corresponding scan-direction angular displacement is

$$\Delta\theta = \mathbf{v} \tau. \quad (\text{SCI-RTC-EQ-027})$$

For every learned RTC operation, the lifecycle is

$$\Lambda = \text{Learn}(\mathcal{L}; \Theta^{\text{eff}}), \quad \Pi = \text{Resolve}(\Lambda, \Theta^{\text{policy}}), \quad \mathbf{y} = \mathcal{A}_\Pi(\mathbf{x}), \quad \frac{\partial \Pi}{\partial \mathbf{j}} = 0 \text{ during apply.} \quad (\text{SCI-RTC-EQ-028})$$

The final equality denotes plan immutability over apply slots; it is not an assumption that the realized operator is stationary at masks, edges, state boundaries, or other declared exceptions.

4 Bounded assumptions and non-goals

SCI-RTC-ASM-001 – Capability.

V0.1 treats Stokes-I primary **xs**. Enabled polarimetry and measured R-channel execution are excluded until separately approved.

SCI-RTC-ASM-002 – Aligned-grid boundary.

The input grid, cadence, phase, mapping, gaps, synthesis/origin state, support, and scan partition are admitted ALIGN facts. Physical event timing, absolute timing accuracy, and a producer epoch may remain unavailable.

SCI-RTC-ASM-003 – Coordinate boundary.

Coordinate-dependent operations assume an admitted AST identity, frame, topology, detector binding, support, validity, and uncertainty where required. The absence of such authority does not license inference or clamping.

SCI-RTC-ASM-004 – Calibration boundary.

Calibration-factor production, atmosphere lineage, beam meaning, uncertainty, and promotion belong to SCI-BEAM/SCI-CAL. RTC conditions on an admitted operator only for a role that authorizes it.

SCI-RTC-ASM-005 – Selected-policy boundary.

Despike, donor, mask, FIR, IIR, notch, state, edge, non-finite, recovery, sampling, diagnostic, and failure choices are selected versioned policies. No current default or numerical threshold is promoted to a universal scientific constant by this contract.

SCI-RTC-ASM-006 – Conditional linearity.

Linear response and covariance identities condition on fixed realized selection and state. Data-derived selectors make the unconditional operation generally nonlinear and nonstationary.

SCI-RTC-ASM-007 – Statistical input.

Moments or covariance are claimed only from a supplied statistical model with declared correlation and limitations. Absence does not imply diagonal, white, stationary, independent, or zero covariance.

SCI-RTC-ASM-008 – Learned-mode scope.

Every learned-then-fixed RTC operation uses one declared learning population, one resolved immutable plan, and one explicit compatibility domain. The first learned sampling scope produces one common observation plan. Per-array, per-scan, heterogeneous-cadence, noise-aware, and continuously adaptive sampling execution are successor scopes.

SCI-RTC-ASM-009 – Analytical authority.

Exact analytical source-times-realized-filter transfer and phase-zero alias accounting define learned-plan admission. Deterministic vectors may falsify a realization, while source injections and observational studies are required for separately claimed performance or science-impact evidence.

SCI-RTC-ASM-010 – Compact representation.

Dense observation-sized response, covariance, or per-sample provenance is not required when a factorized, sparse, interval-based, or generative representation exactly reconstructs the operator, support, causes, and exceptions.

SCI-RTC-ASM-011 – Downstream authority.

RTC supplies causal inputs but does not own PTC cleaning/weights, VAL eligibility policy, MAP estimation, BEAM inference, empirical NOI products, MAP-003 response tracing, map filtering, source fitting, or fruit-loop recurrence.

SCI-RTC-ASM-012 – Claim separation.

The contract is an algebraic and semantic authority only. It contains no claim that a present implementation conforms, that a representation is faithful, that observational performance is adequate, or that a mode is ready for production.

4.1 Explicit non-goals

This revision does not derive ALIGN timing, AST coordinates, SCI-BEAM or SCI-CAL calibration, PTC, VAL, MAP, or FLT estimators, source fitting, or recurrence; authorize absolute-timing or astrometric corrections; re-design mature despiking/FIR/IIR/notch numerics; choose universal numerical thresholds; authorize online adaptation; activate polarimetry or measured R; prescribe source classes, files, storage schemas, or product names; authorize per-array/per-scan cadence; inspect implementation; perform validation or reductions; or assert production status.

5 Sequential normative requirements

The following identifiers are stable within v0.1/r0.2. Conformance is requirement-by-requirement; satisfying a neighboring requirement does not waive another one.

Area	ID	Normative statement
Boundary	SCI-RTC-REQ-001	RTC shall admit only an aligned primary xs stream whose signal meaning, ordered identity, assigned grid, support, origin/synthesis state, and validity are declared.
Boundary	SCI-RTC-REQ-002	Every requested product role shall declare physical quantity, unit, sign, reference/baseline, valid domain, and required consumers; no channel or role shall inherit another role's meaning.
Boundary	SCI-RTC-REQ-003	The primary standardized Beammap detector signal shall remain raw $\Delta f/f$ and shall not be silently replaced by a calibrated timestream or map.
Boundary	SCI-RTC-REQ-004	A calibrated mJy/beam domain shall be produced only for a separately CAL-authorized role with admitted factor identity, convention, atmosphere lineage, detector binding, validity, uncertainty availability, and provenance.
Boundary	SCI-RTC-REQ-005	RTC shall own the exact ordered application and response consequence of an imported CAL operator but shall not derive, repair, promote, or reinterpret its factor science.
Identity	SCI-RTC-REQ-006	Observation, Tune, array, network/interface, detector occurrence, APT binding, native row, aligned slot, scan, science interval, context interval, output slot, stage, and product identity shall remain distinct and immutable.
Identity	SCI-RTC-REQ-007	The ordered detector and sample shapes, index bases, native-to-aligned mapping, aligned-to-output map, scan membership, and memory order shall be explicit; indices shall not serve as cross-artifact detector identity.
Identity	SCI-RTC-REQ-008	RTC shall preserve the admitted ALIGN cadence, phase, rate, gaps, support, and synthesis/origin state without claiming physical integration-event timing or absolute epoch authority.
Identity	SCI-RTC-REQ-009	A coordinate-dependent operation shall consume the exact AST identity, frame, topology, detector binding, support, and validity required by its product role, with no implicit frame conversion.
State	SCI-RTC-REQ-010	Requested, effective, observation-resolved, learned/resolved when applicable, and realized state shall be recorded separately in one-way order; later state shall not overwrite the accepted request.
State	SCI-RTC-REQ-011	Outer, inner, full, mini, diagnostic, simulated, processed, and any other produced stage roles shall have distinct immutable identities and explicit parent/processing links without requiring duplicate computation.
Policy	SCI-RTC-REQ-012	Every desprike, donor, replacement, mask, filter, state, edge, non-finite, recovery, sampling, flag, diagnostic, and fallback action shall be governed by an admitted versioned selected policy rather than an undocumented default.
Order	SCI-RTC-REQ-013	For a CAL-authorized role, calibration shall precede cross-detector replacement unless complete affine equivalence in equation (SCI-RTC-EQ-004) is proved for the exact unit, factor, offset, response, uncertainty, identity, validity, and domain.

Area	ID	Normative statement
Donor	SCI-RTC-REQ-014	A raw donor q to raw target d using the multiplicative convention $z_i = \phi_i x_i$ shall use ϕ_q/ϕ_d only when both flxscale factors are valid for the exact occurrences under the same compatible convention/domain and $\phi_d \neq 0$.
Donor	SCI-RTC-REQ-015	The raw donor route in REQ-014 shall use flxscale and shall neither require nor restore a scientific role for legacy responsivity .
Donor	SCI-RTC-REQ-016	When REQ-014 is not available, replacement shall fail under that route unless a separate donor-to-target authority declares direction, quantity/unit domain, support, uncertainty, validity, and provenance.
Donor	SCI-RTC-REQ-017	The despiking detector, target indicator, candidate eligibility, topology, score, tie rule, donor weights, time relation, source-mask/flag use, fallback, and every realized donor link shall be reconstructible.
Donor	SCI-RTC-REQ-018	A replaced target shall retain target identity, link every donor and factor, preserve the replacement cause, and expose correlations from reused or shared donors; finiteness shall not create independent exposure.
Influence	SCI-RTC-REQ-019	Every output transitively influenced by an ALIGN-synthesized cell shall be scientifically ineligible while preserving its numerical effect, typed cause, and support.
Influence	SCI-RTC-REQ-020	Every output transitively influenced by an RTC-replaced cell shall be scientifically ineligible while preserving its numerical effect, typed cause, donor linkage, and support.
Filter	SCI-RTC-REQ-021	Every enabled response-changing stage shall record chronological order, exact full-precision coefficients or exact digest, coefficient convention, normalization, numeric precision, rate, direction, state, context, mask, edge, non-finite policy, and exceptions once per coherent segment.
Filter	SCI-RTC-REQ-022	An FIR claim shall expose exact impulse support, DC gain, phase convention, and normalization; a label such as “normalized” shall not substitute for the coefficient sum or response.
Filter	SCI-RTC-REQ-023	An IIR or notch claim shall expose an exact recurrence or state-space equivalent, realized poles/zeros where applicable, initial/final state, reset/carry boundary, transient/guard rule, precision, and stability status.
Mask	SCI-RTC-REQ-024	A processing mask shall be represented as an operator control, including geometry or selection, boundary convention, dilation, denominator/segment behavior, and state interaction; it shall not be treated as acquisition validity or confidence.
Mask	SCI-RTC-REQ-025	Invalid, unavailable, wrong-frame, out-of-domain, or ambiguously bound coordinates shall not become outside-source samples and shall fail the affected required coordinate operation.
Finite	SCI-RTC-REQ-026	A non-finite required sample, factor, coefficient, coordinate, or filter state shall be rejected, marked unavailable over its exact transitive influence, or handled by an explicitly authorized prior replacement/recovery rule; it shall not be silently coerced to zero.
Edge	SCI-RTC-REQ-027	Boundary extension, incomplete context, filter guards, IIR recovery, short/empty scans, and output selection shall be explicit; an edge or incomplete-state cell shall not be labeled an ordinary interior sample.

Area	ID	Normative statement
Sampling	SCI-RTC-REQ-028	V0.1 downsampling shall use only phase-zero point selection $y[n] = u[Mn]$; arithmetic-mean and other block-aggregate downsampling are excluded.
Sampling	SCI-RTC-REQ-029	Each coherent segment shall record M , zero phase, input/output rates and grids, the cardinality of equation (SCI-RTC-EQ-011), selected-point representative time, full transitive support, edge disposition, and flag/influence aggregation.
Sampling	SCI-RTC-REQ-030	Any alias-free or bounded-alias claim shall include the exact realized prefilter and all folded bands of equation (SCI-RTC-EQ-014), with a declared input band, metric, bound, and unavailable state; point selection alone shall not be called anti-alias filtering.
Sampling	SCI-RTC-REQ-031	Fixed mode shall resolve an admitted factor/filter/rate/compatibility plan before application and record it independently of realized output.
Learned	SCI-RTC-REQ-032	Every learned RTC operation shall preserve distinct requested, effective, observation-resolved, learned-evidence, resolved-plan, and applied states and shall not advertise a plan before admitted evidence and numeric policy are complete.
Learned	SCI-RTC-REQ-033	Learned resolution shall choose the largest admitted integer factor only after astronomical-transfer, exact phase-zero alias, sampling, filter-realizability, and downstream-compatibility constraints all pass using the smallest admitted beam and maximum valid in-scan speed.
Learned	SCI-RTC-REQ-034	The first learned applied scope shall use one common immutable observation cadence and filter across arrays and scans; percentile speed shall remain diagnostic rather than safety authority.
Learned	SCI-RTC-REQ-035	Apply shall consume exactly one immutable resolved plan and shall not retune, adapt, or silently fall back; an unavailable or rejected plan shall follow the selected failure/fallback policy before apply begins.
Learned	SCI-RTC-REQ-036	Restart shall restore a state-complete resolved plan, verify exact plan/input compatibility, and reject stale or incompatible beam, speed, scan, coefficient, rate, phase, role, or parent identity.
Response	SCI-RTC-REQ-037	The atomic bundle shall contain the complete realized response of SCI-RTC-DEF-010 or a typed unavailable state; a partial filter kernel shall not be labeled the conditioned response.
Response	SCI-RTC-REQ-038	Selection, detector mixing, time-varying calibration, masking, finite boundaries, state, or multirate behavior shall use a local/factorized detector-time response or truthful unavailability rather than an unjustified scalar transfer.
Response	SCI-RTC-REQ-039	A scalar LTI response shall be claimed only on a stated uniform, fixed-state, detector-separable, translation-invariant interior domain with no data-derived selector or undeclared rate change.
Response	SCI-RTC-REQ-040	Where defined, response shall distinguish amplitude, power, phase, group delay, sampling rate, Fourier convention, alias images, support, and response uncertainty; undefined phase/delay at a response zero shall remain unavailable.
Response	SCI-RTC-REQ-041	Output support shall be expanded transitively through alignment, replacement, every filter, state boundary, edge rule, and phase-zero selection; a selected center index shall not stand in for full support.

Area	ID	Normative statement
Statistics	SCI-RTC-REQ-042	Conditional mean and covariance shall follow equations (SCI-RTC-EQ-016a)–(SCI-RTC-EQ-016b) only for an admitted input model and fixed realized state, with reused donors and overlapping support represented as correlations.
Statistics	SCI-RTC-REQ-043	When selection depends on modeled random data, selection bias and selection covariance shall be modeled or explicitly unavailable; they shall not be set to zero merely by conditioning on the realized selector.
Statistics	SCI-RTC-REQ-044	Calibration, atmosphere, beam, donor-scale, coefficient, cadence/phase, response, and model nuisance terms shall retain value, validity, uncertainty, provenance, and correlation scope, including required cross terms.
Statistics	SCI-RTC-REQ-045	Unknown covariance shall remain unavailable and shall not be replaced by diagonal, white, stationary, independent, zero, or scalar-weight assumptions; conditional inverse variance shall not be called full precision or total uncertainty.
Validity	SCI-RTC-REQ-046	Acquisition support, original validity, ALIGN synthesis, numerical computability, coordinate validity, flags/causes, operator masks, replacement, transitive influence, complete support, response status, consumer eligibility, uncertainty availability, and provenance shall remain separate states.
Validity	SCI-RTC-REQ-047	A flag shall describe a cause and shall not automatically become a mask, invalidity, weight, or universal eligibility decision; aggregation shall preserve every contributing cause required by the selected policy.
Output	SCI-RTC-REQ-048	RTC shall produce the role-specific atomic bundle of equation (SCI-RTC-EQ-022); conditioned TOD alone shall not be reported as a complete required RTC product.
Output	SCI-RTC-REQ-049	Required malformed identity/state or missing authority shall propagate to atomic required-output failure; optional unavailable provenance detail shall be typed and numerically inert.
Output	SCI-RTC-REQ-050	Compact provenance shall reconstruct input/parent identity, detector binding, grid and intervals, calibration state, donor links, ordered filters, masks, coefficients/state/edges, sampling, response/uncertainty status, exceptions, counts, and all one-way state records.
Output	SCI-RTC-REQ-051	Each diagnostic shall name estimator, unit, support, mask/selection, validity, availability, and whether it is inert, advisory, or enters the selected policy; optional diagnostic detail shall not change values, flags, eligibility, cardinality, or failure.
Consumer	SCI-RTC-REQ-052	RTC shall supply cause-preserving validity inputs rather than a universal downstream eligibility bit; each consumer shall bind the exact RTC parent and shall not strengthen unavailable timing, coordinate, calibration, response, covariance, or eligibility claims.
Consumer	SCI-RTC-REQ-053	PTC-disabled terminal RTC output shall be a valid requested state when selected; no PTC or map product shall be implied, while any requested downstream product shall require the exact admitted RTC bundle.
Claims	SCI-RTC-REQ-054	Algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, and production readiness shall be stated and assessed as separate claims.

Area	ID	Normative statement
Lifecycle	SCI-RTC-REQ-055	Every learned operation shall name its learning population, estimator, support, selection, exclusions, quality, uncertainty, validity, and compatibility domain; same-observation learning shall be identified as data-dependent.
Lifecycle	SCI-RTC-REQ-056	Resolve shall convert evidence and fixed policy into one immutable plan containing every applicable sequence, family, coefficient identity, rate/convention, order/taps, band/notch parameter, applicability, mask, donor, edge/guard/reset, decimation/timing, response/support, uncertainty, validity, and downstream-compatibility field.
Lifecycle	SCI-RTC-REQ-057	A plan shall be resolved only when every required predicate and tolerance passes; missing evidence, missing tolerance, rejected evidence, or unavailable compatibility shall not be treated as pass.
Lifecycle	SCI-RTC-REQ-058	Any coefficient or operator parameter that evolves during apply shall be governed by a separately authorized online-adaptive estimator contract; it shall not be represented as learned-then-fixed filtering.
Design	SCI-RTC-REQ-059	Every despoke, donor, mask, notch, low/high/band-pass, FIR, IIR, decimation, edge/state, or non-finite operation shall declare scientific purpose, signal/contaminant model, mathematical operator, design parameters, learn/resolve/apply behavior, astronomical and calibration consequences, validation/falsification, and unavailable states.
Design	SCI-RTC-REQ-060	Filter admission shall use the exact source-times-realized-response over the declared scan-speed/direction, projected-beam, source-class, prior-response, and cadence domain; $f \sim v/\theta$, a scalar FWHM, or a nominal cutoff shall remain pedagogical only.
Notch	SCI-RTC-REQ-061	Every notch shall resolve center, rate/frequency convention, width or Q, depth, family/order, poles/zeros where applicable, phase/group delay, state, transient/guard, applicability, learned uncertainty, drift tolerance, and combined response for overlaps.
Bands	SCI-RTC-REQ-062	Every admitted low-, high-, or band-pass shall resolve pass/transition/stop bands, ripple, attenuation, phase/delay, family, order/taps, precision, edges/guards, retained astronomical modes, and unavailable modes; no class label shall substitute for exact response.

Area	ID	Normative statement
FIR	SCI-RTC-REQ-063	FIR order or tap count shall follow owner-resolved transfer, alias, phase, support, guard, precision, and realizability constraints such as equation (SCI-RTC-EQ-025); an arbitrary integer or approximate family formula shall not be scientific authority.
Donor	SCI-RTC-REQ-064	Donor replacement shall be identified as a continuity surrogate at the target identity and shall separately establish numerical convention transfer, donor admissibility, and scientific eligibility; compatible <code>flxscale</code> shall not be evidence of common sky signal.
Donor	SCI-RTC-REQ-065	Beammap-role donor scaling shall use an explicitly identified prior APT or other pre-existing calibration authority and shall not use a new <code>flxscale</code> derived from the same Beammap conditioning lineage.
Sampling	SCI-RTC-REQ-066	Decimation factor and prefilter shall be justified from cadence, exact response, folded-band alias bounds, smallest projected crossing time, maximum valid scan speed, source class, timing/coordinate accuracy, and downstream compatibility.
Timing	SCI-RTC-REQ-067	Each plan shall resolve temporal registration through zero phase, shifted output time, coordinate correction, response-aware downstream use, or an approved bound on residual centroid/PSF effect; selected grid time alone shall not imply response centering.
CAL	SCI-RTC-REQ-068	Calibration transfer between Beammap and target processing shall bind the complete RTC plans and require matching response or explicit calculation, renormalization, or observational evidence; identical operation labels or preserved units are insufficient.
Validation	SCI-RTC-REQ-069	Validation of a notch, broad-band/FIR, donor, or decimation plan shall vary its scientific design parameters over the admitted scan/beam/source domain and measure astronomical transfer, phase/centroid, PSF/morphology, ringing, support/edge loss, covariance, and alias or interference residual as applicable.
Claims	SCI-RTC-REQ-070	No plan shall be called science-qualified solely because coefficient algebra, optimization, serialization, or deterministic operator fixtures pass.

6 Falsifiable predictions and limiting cases

Each prediction is an algebraic or semantic falsification target. A future conformance activity may choose an exact numerical tolerance justified by the operation and representation before observing candidate output. No result is reported by this document.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-001	Identity chain: raw role, $A = I$, no replacement or filter, $M = 1$, valid original samples. Values, unit, order, identities, cardinality, response, flags, and support are unchanged.	Any hidden calibration, value/order/cardinality change, flag loss, nonidentity kernel, or new claim of timing/coordinate authority.
SCI-RTC-PRED-002	Role separation: feed the same dimensionless raw Beammap input to raw and explicitly CAL-authorized requests. The raw output remains $\Delta f/f$; the calibrated output exists only with its admitted CAL state.	A raw product labeled mJy/beam, a calibrated product produced without CAL authority, or silent interchange between the products.
SCI-RTC-PRED-003	Raw donor direction: choose $\phi_q = 2$, $\phi_d = 4$, and donor $x_q = 3$. Equation (SCI-RTC-EQ-002) predicts raw target replacement 1.5 and equal calibrated amplitudes 6.	Use of ϕ_d/ϕ_q , output other than 1.5, unequal calibrated amplitude, or dependence on legacy responsivity .
SCI-RTC-PRED-004	Raw donor availability: independently make either factor invalid, convention/domain-incompatible, occurrence-mismatched, or set $\phi_d = 0$. The flxscale route is unavailable.	Finite replacement, stale factor reuse, row-number-only binding, zero insertion, or success without separate donor authority.
SCI-RTC-PRED-005	Order: compare calibration-before-replacement with a proposed reversed affine path using unequal factors and nonzero offsets. Outputs and local response agree only if equation (SCI-RTC-EQ-004) holds completely.	Equality asserted from multiplicative factors alone, ignored offsets/uncertainty, or fixture equality labeled a general proof.
SCI-RTC-PRED-006	Interior target/donor impulses under fixed selection. Each output column equals equation (SCI-RTC-EQ-012); donor energy appears in the declared detector column, support, scale, and phase.	Missing or circularly wrapped energy, unexplained mixing, wrong donor identity/direction, support, normalization, or phase.
SCI-RTC-PRED-007	Constant input with sufficient context. Output equals the constant times realized DC gain after the declared transient; a DC-preserving chain is exactly constant.	Drift, divide-by-zero mask normalization, edge values advertised as interior, or a false DC-preservation claim.
SCI-RTC-PRED-008	Interior step with complete context. Response equals the cumulative impulse response and settles according to DC gain and stability.	Undeclared overshoot, failure to settle, hidden state leakage, or disagreement with the impulse-derived response.
SCI-RTC-PRED-009	Affine ramp through each FIR and admitted alignment mapping. Output slope and offset follow the zeroth and first moments of the exact kernel.	Unexplained clipping, wrong phase/slope/offset, or a ramp-preservation claim without the required moments.
SCI-RTC-PRED-010	Long bin-centered and off-bin sinusoids at DC, passband interior/edge, transition, stopband, and near output Nyquist. Amplitude and phase follow equations (SCI-RTC-EQ-013) and (SCI-RTC-EQ-014).	Gain/phase outside preregistered tolerance, unrecorded frequency convention, or an aliased component labeled passband signal.
SCI-RTC-PRED-011	Sinusoids at the realized notch center and symmetric offsets under the exact rate and state. Depth, width, phase, and transient follow realized zeros/poles.	Center/depth/width mismatch, unstable pole, rate mismatch, hidden state, or an edge transient reported ordinary-valid.
SCI-RTC-PRED-012	Coordinate mask with valid inside/outside values plus invalid, unavailable, wrong-frame, out-of-domain, and ambiguously bound coordinates. Invalid classes fail the affected operation and remain distinct from outside.	Invalid becoming outside/unmasked, entering geometry, clamping to a center/edge, or loss of the distinct cause.
SCI-RTC-PRED-013	Samples just inside, exactly on, and just outside each selected mask boundary and temporal dilation. Membership follows the recorded inclusive/exclusive and dilation policy identically across execution modes.	An undocumented boundary, frame mismatch, asymmetric dilation, or sequence-dependent membership.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-014	Impulses and constants at first/last support cells and immediately inside complete FIR context. Values, guard, support, and output admission follow the selected edge rule.	Undeclared padding, truncation, wrap, renormalization, guard, or output selection.
SCI-RTC-PRED-015	Zero, impulse, and constant inputs split and rejoined at scan/chunk/state boundaries for an IIR. Results follow the declared reset/carry and terminal-state policy.	Prior-observation leakage, hidden carry/reset, unbounded non-finite poisoning, or contradiction between split and declared state semantics.
SCI-RTC-PRED-016	Lengths $0, 1, L - 1, L, L + 1$ around every support/guard threshold, with stable scan identity.	Crash, underflow, stale output, silent scan deletion/renumbering, or ordinary-valid status without required context.
SCI-RTC-PRED-017	Inject NaN, $+\infty$, and $-\infty$ separately into a sample, factor, coordinate, coefficient, and IIR state. Failure or unavailable influence follows REQ-026 exactly.	Silent zero/coercion, finite-looking output without cause, stale state recovery, or an unbounded tail reported valid.
SCI-RTC-PRED-018	Place one typed cause at every member of a phase-zero output's full support, including a noncenter synthesized or replaced cell. Each cause propagates and either such influence makes the output scientifically ineligible.	Center-only aggregation, erased cause, finite output called eligible, or replacement/synthesis treated as independent exposure.
SCI-RTC-PRED-019	For $M = 1, 2, 3, 4, 5$ and $N = 0$ through several incomplete final strides, enumerate $j = Mn$. Cardinality and selected time follow equation (SCI-RTC-EQ-011); no block mean appears.	Wrong N_y , dropped/duplicated point, nonzero phase, averaged value, timestamp misidentification, or rate mismatch.
SCI-RTC-PRED-020	Two sinusoids separated by an integer multiple of f_{out} , before and after the realized prefilter. Folded amplitude is the sum in equation (SCI-RTC-EQ-014).	A folded term missing from response, inadequate stopband labeled alias-free, or incorrect image frequency/amplitude.
SCI-RTC-PRED-021	Reuse one donor for two targets and give neighboring outputs overlapping filter support. Equation (SCI-RTC-EQ-016b) predicts cross-target and temporal covariance when the admitted input covariance permits it.	Independent weights assigned, off-diagonal terms silently dropped, or reused donor described as additional exposure.
SCI-RTC-PRED-022	Enable a response-changing mask, donor mix, stateful filter, edge, and decimator while providing only a scalar FIR transfer. Complete response state is unavailable.	The partial scalar transfer labeled complete, or downstream use that assumes omitted stages are identity.
SCI-RTC-PRED-023	Learned candidates include at least two safe factors and one failing each astronomical-transfer, alias, sampling, realizability, and compatibility constraint. Equation (SCI-RTC-EQ-021) chooses the largest passing factor.	A smaller safe factor selected without rule, any failing factor applied, a percentile speed used for safety, or a missing numeric predicate treated as pass.
SCI-RTC-PRED-024	Serialize and restart a learned plan, then alter one of beam identity, speed support, scan set, coefficient precision, rate, phase, role, or parent. Apply rejects the mismatch.	Silent relearning, retuning, native fallback, stale-plan application, or realized state rewriting the request.
SCI-RTC-PRED-025	Execute two observations sequentially with distinct rates, constants, factors, masks, donors, states, and output parents.	Any coefficient, state, factor, donor, mask, rate, flag, or product link leaks from the first observation.
SCI-RTC-PRED-026	Toggle optional detailed provenance and inert diagnostics while holding the selected policy and realized operator fixed. Values, flags, influence, eligibility, cardinality, response, and failures are identical.	Any scientific or numerical result changes, or an observational/production claim appears from the toggle.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-027	Bind identical learned parameters first to a separate training population and then to the apply observation. The numerical fixed-state response can match, but the second plan records data dependence and parameter/selection uncertainty.	Same-observation learning represented as fixed a priori, or conditional linearity used to erase selection or parameter uncertainty.
SCI-RTC-PRED-028	Remove in turn one required learning-population identity, tolerance, uncertainty, or downstream-compatibility predicate before resolve. The dependent plan remains unresolved and apply does not begin.	Missing evidence treated as pass, a default inserted, or output represented as applied learned mode.
SCI-RTC-PRED-029	Serialize one resolved plan and attempt during apply to alter a notch center, cutoff, tap count, donor rule, or decimation factor. The mutation is rejected or belongs to a separately authorized adaptive estimator.	Hidden retuning, silent fallback, or a realized record that rewrites the resolved plan.
SCI-RTC-PRED-030	Sweep notch center error, line drift, width, depth, Q/order, initialization, direction, and overlap. Residual line and astronomical amplitude, phase, centroid, covariance, transient, and edge loss follow the exact combined response.	Only line disappearance measured, or any changed astronomical observable omitted from the plan's validity evidence.
SCI-RTC-PRED-031	Scan the same projected source at multiple speeds and directions through a fixed filter. Its temporal spectrum and retained amplitude/PSF change according to equation (SCI-RTC-EQ-023) and exact response.	One nominal cutoff or scalar beam treated as speed- and direction-independent proof of preservation.
SCI-RTC-PRED-032	Design FIR candidates over multiple tap counts and coefficient precisions. The selected N_* is the smallest candidate passing every registered transfer, alias, phase, support, guard, and realizability predicate.	Arbitrary configured taps, approximate order formula promoted to authority, or a candidate with any failed predicate selected.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-033	Inject compact and extended sources through admitted high- or band-pass candidates. Low-frequency rejection produces the predicted lost broad modes, compensated response, or negative lobes.	Unit preservation labeled flux preservation, or removed absolute/extended modes reported available without response restoration evidence.
SCI-RTC-PRED-034	Choose donor and target with compatible <code>flxscale</code> but deliberately different sky signal or beam coupling. Raw-domain convention transfer remains algebraically correct while the surrogate is not a missing-sky reconstruction.	Compatible factors claimed to prove physical donor equivalence or independent exposure.
SCI-RTC-PRED-035	Attempt Beammap donor scaling with a new <code>flxscale</code> derived from that same conditioned Beammap. The route is rejected until an identified prior factor authority is supplied.	Circular use of the new factor or absence of the prior authority in lineage.
SCI-RTC-PRED-036	Apply filters with known nonzero group delay to Pointing/Beammap scans in opposite directions. Centroid displacement follows equation (SCI-RTC-EQ-027) with opposite scan-direction sign unless time/coordinates are corrected.	Selected grid time treated as response center, wrong displacement sign, or unrecorded correction.
SCI-RTC-PRED-037	Process calibrator and target with plans that share a nominal label but differ in coefficients, rate, phase, state, edge, or decimation. Cross-plan calibration transfer is unavailable absent explicit response accounting or observational evidence.	Nominal label, preserved unit, or <code>flxscale</code> alone treated as RTC-plan compatibility.
SCI-RTC-PRED-038	Pass all deterministic coefficient and operator fixtures but withhold the required source/scan design studies. Algebraic correctness may be supported; science qualification remains unassessed.	Deterministic pass promoted to observational performance, science qualification, or production readiness.

6.1 Analytic limits

The following limits are required: a compatible donor pair with $\phi_q = \phi_d$ gives unit raw donor scale; a replacement indicator tending to zero gives identity; filter coefficients tending to a Kronecker delta give identity; $M \rightarrow 1$ gives no rate change; a source-mask radius tending to zero selects only exact valid coincidences under the declared boundary; complete context approaches the declared interior response; and a causal pole approaching the unit circle exposes increasing memory and never retains a stability claim at or outside the circle.

End of shared normative core: SCI-RTC-v0.1/r0.2. View-specific material cannot add to or modify this scientific authority.

A Informative conformance-record schema

This appendix supplies bookkeeping guidance only. It does not add a requirement. A review record should contain the following fields.

Field	Evidence content
Record identity	Stable review ID, date, reviewer role, exact contract revision, and supersession status.
Claim layer	Exactly one primary layer: implementation conformity, representation fidelity, observational performance, science impact, or production readiness.
Target identity	Exact source/build/artifact identity and any configuration or runtime identity needed to reproduce the observation.
Authority binding	One or more SCI-RTC requirement or prediction IDs; an equation/definition may be cited only through the shared core.
Preconditions	Product role, signal domain, detector occurrences, grid, intervals, requested/effective/resolved state, and owner decisions required by the check.
Method	Static trace, deterministic vector, bounded matrix calculation, serialization inspection, failure injection, or other named method.
Expected result	Exact invariant and a tolerance justified before observing candidate output.
Observed result	Values, states, messages, artifacts, and exceptions without interpretation hidden in a generic pass flag.
Disposition	Pass, fail, not exercised, blocked by open owner decision, or evidence unavailable.
Claim limit	Explicit statement of what the record does not establish.

No status may be inferred from an omitted record. “Not exercised” means the evidence was not attempted; “blocked” means a named unresolved authority prevents a meaningful check; and “unavailable” means the attempted evidence cannot support the claim. These dispositions are not passes.

B Informative requirement-review routing

The table suggests evidence classes without prescribing implementation structure. Reviewers should use the exact crosswalk shipped with the package for one-to-one coverage.

Requirement range	Primary review surface	Useful evidence classes
REQ-001–005	Boundary and signal role	Typed input/output inspection, unit/domain failure injection, imported-authority lineage trace.
REQ-006–012	Identity, state, and selected policy	Serialization round trip, distinct-identity fixtures, request-to-realized trace, unsupported-policy rejection.
REQ-013–018	Ordering and donor transfer	Symbolic/static order trace, forced donor vectors, invalid-factor matrix, donor reuse/tie/fallback fixtures.
REQ-019–020	Transitive influence	Noncenter synthesis/replacement injection, full-support cause propagation, downstream eligibility handoff inspection.
REQ-021–027	Filters, masks, edges, and finite policy	Coefficient/state serialization, impulse/step/constant/ramp/notch vectors, mask-boundary and non-finite failure injection.
REQ-028–031	Fixed phase-zero sampling	Enumerated length/factor cases, selected index/time identity, rate trace, exact folded-band calculation.
REQ-032–036	Learned plan lifecycle	Candidate decision table, maximum-safe selection, common-plan identity, immutable apply and restart mismatch fixtures.
REQ-037–041	Complete response and support	Factorization reconstruction, local Jacobian fixtures, LTI-domain proof, edge/state/support expansion.
REQ-042–045	Statistical availability	Bounded covariance matrices, reused-donor correlation fixture, selector/nuisance availability inspection.
REQ-046–051	Validity and atomic output	Orthogonal-state fixtures, atomic write/failure injection, provenance reconstruction, inert-diagnostic toggle.
REQ-052–054	Consumers and claims	Parent-binding inspection, unavailable-state preservation, disabled-PTC terminal path, claim-label audit.
REQ-055–058	Learn–resolve–apply lifecycle	Learning-population identity, predicate-completeness matrix, immutable-plan trace, mutation rejection, adaptive-estimator separation.
REQ-059–063	Scientific filter design	Purpose/model register, scan-beam transfer studies, notch and band specifications, constrained tap/order selection.
REQ-064–068	Donor, decimation, timing, calibration	Donor-meaning separation, circular-factor rejection, folded-band calculation, centroid/coordinate tests, cross-plan response comparison.
REQ-069–070	Validation and claims	Preregistered design sweeps and strict claim-layer disposition.

C Informative response and representation checks

Review the response at the same granularity as the realized operation. A stored coefficient vector can support a component-level check; it does not by itself establish complete response. For each coherent segment, trace the ordered factorization, exact coefficients and precision, mask/state/edge exceptions, detector mixing, sampling phase, and support expansion to the artifact presented to consumers. Mark complete response unavailable if any enabled response-changing stage cannot be represented truthfully.

Use the prediction IDs rather than inventing new expected science after observing output. Interior impulses and sinusoids exercise fixed linear components; donor impulses exercise detector mixing; split-state fixtures exercise recurrence; noncenter causes exercise transitive influence; and odd/even factor enumerations exercise phase-zero identity. These checks are complementary. An interior transfer match cannot close edges or selection, and an output-value match cannot close identity, response, or provenance.

Representation fidelity is a separate record. Reconstruct the exact operator and exceptions from persisted factors, intervals, mappings, and state. Verify that omitted detail is typed optional and inert. A readable plot or plausible header is not proof that the stored representation is complete.

D Informative state and failure checks

Exercise each semantic state independently: not requested, disabled, automatic, rejected, invalid, failed, unavailable, unsupported, and omitted. Also separate original validity, synthesis, replacement, mask membership, numerical finiteness, edge/context, response availability, eligibility, and uncertainty. A fixture that combines every failure into NaN or one generic bit cannot demonstrate these distinctions.

For each required output member, inject a bounded write or authority failure and verify atomic completion behavior. Inspect that no prior-observation factor, donor, state, mask, rate, product link, or completion marker remains. Then toggle optional detailed provenance and inert diagnostics; the scientific and numerical bundle must remain identical apart from the permitted optional detail.

Every learned operation requires an additional stop check before apply. The review must bind the declared learning population and every numerical predicate to approved authority, show the complete candidate disposition, and preserve learned-evidence and resolved-plan state. If the resolved plan is absent or incompatible, apply does not begin. A native-cadence result produced after silent fallback is not evidence of an applied learned plan.

E Informative completion checklist

- Every SCI-RTC-REQ identifier has one explicit disposition and evidence record; ranges are not used as a substitute for per-ID closure.
- Every applicable SCI-RTC-PRED identifier has a preregistered expected result and tolerance, plus observed values and messages.
- Every open owner decision is either resolved by approved authority or blocks the named operation/claim exactly as the ledger states.
- Both raw Beammap and separately CAL-authorized role boundaries are exercised without cross-role inheritance.
- The owner-modified raw donor ratio is checked in both direction and availability cases, with no dependency on legacy responsivity.
- Full-support synthesis/replacement influence is checked at noncenter cells and remains scientifically ineligible downstream.
- Complete response includes every enabled stage or is typed unavailable.
- Requested, effective, observation-resolved, learned-evidence, resolved-plan, realized, and immutable product/stage identities remain distinct.
- Each learned operation names its learning population, and apply rejects any mutation of the resolved plan; online adaptation is a separate unavailable or separately authorized estimator.
- Filter validation varies scientific design choices over the admitted scan, projected-beam, and source domain; coefficient fixtures alone do not establish science qualification.
- Required failures are atomic; optional detail is inert; unexpected error-level messages are zero in any run claimed successful.
- The final report names exactly one supported claim layer and explicitly withholds all stronger layers.

This checklist does not itself establish any item. It is a review index for future evidence governed by the shared core.